Coupled Half-Bridge *LLC* Resonant Converter With Reduced Switches and Current Sharing Ability

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***Abstract*—A novel coupled half-bridge *LLC* resonant converter is proposed in this article, which has fewer switches and automatic current sharing ability compared with the traditional parallel half-bridge *LLC* resonant converter. Since the two resonant networks in the proposed converter share the same power switches, the number of devices is reduced by half, which is beneficial for increasing system power density. In addition, the resonant inductor is shared by the two resonant networks, which means that the uneven current in the two *LLC* resonant converters, caused by differences in resonant parameters, can be shared. The operation mode of the proposed converter is analyzed, and it has the same characteristics as the traditional *LLC* resonant converter. Besides, the coupled resonant network is analyzed on the basis of fundamental harmonic analysis, and the current sharing effect of the proposed converter is analyzed. Further, a 1 kW experimental platform is built, and the theoretical analysis and simulation results are validated by experimental results.**

***Index Terms*—Common inductor, common switch, coupled half-bridge *LLC* resonant converter, current sharing.**

I. Introduction

LLC resonant converter has been widely used in electric vehicles, data center power supplies, communication power supplies and other fields, because of its soft switching, input and output high-frequency isolation characteristics [1], [2], [3], [4], [5]. However, with the increase in power supply level demands, a single *LLC* resonant converter cannot meet the output power requirement due to the limitations of power devices. The parallel systems of *LLC* converters can decrease the current stress in each switch and diode compared with a single *LLC* module at the same power level, as illustrated in Fig. 1. Therefore, parallel systems have gotten more and more attention in high-power applications. However, *LLC* resonant converters are more parameter-sensitive than other converters, and manufacturing errors cause serious current imbalance [27]. Thus, current sharing is a concern, leading to extensive research. Current methods primarily focus on current sharing approaches.

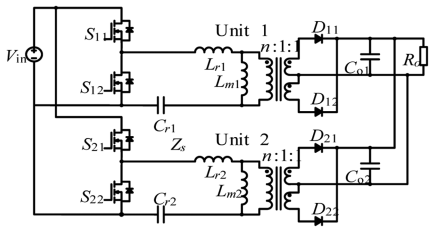


Fig. 1. Traditional two-phase parallel half-bridge *LLC* resonant converter.

A current sharing method for parallel *LLC* resonant converters using the series current equalization property of transformers is proposed in [6], and the proposed method are very simple to implement with no additional components or control strategies; only the secondary windings of transformers need to be grouped. However, this method requires additional windings on the secondary side, which inevitably increases the size of transformers and is not conducive to achieving high power density. [7] has proposed a reliable and efficient magnetic integration technique; the current sharing is achieved by coupling the resonant inductor between the phases. The three-phase circuit topology proposed in [8] effectively suppresses the maximum peak current that may occur in the circuit through a current sharing transformer among the three phases, but it increases the volume of the magnetic components. In addition, many other scholars have conducted many studies on magnetic coupling [9], [10], [11]. Although these methods can achieve automatic current sharing, they also increase the complexity of the process and do not allow for further increases in the power density of coupled *LLC* resonant converters.

A current-sharing scheme for a two-phase *LLC* converter based on virtual controllable voltage sources is proposed in [12]. The main approach involves incorporating the auxiliary winding of the transformer from one phase into the resonant network. Current sharing is achieved through phase-shift modulation, with the phase shift between the two phases naturally maintained. Another method presented in [13], achieves current sharing by adding a small transformer and an auxiliary low-power DC–DC converter to each phase, where the duty cycle

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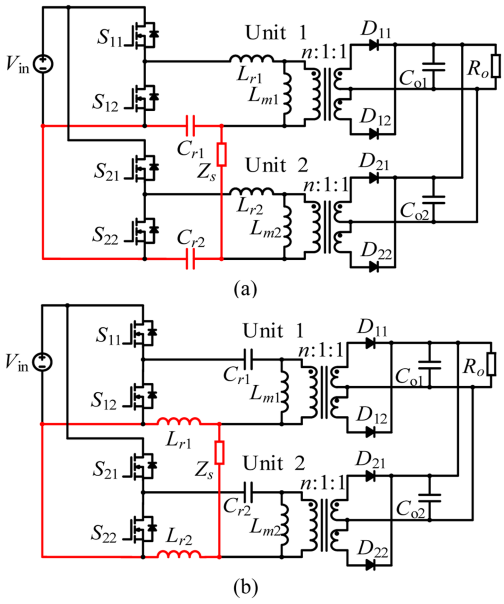


Fig. 2. Common capacitor and common inductor *LLC* resonant converter with current sharing ability. (a) Common capacitor two-phase LLC resonant converter. (b) Common inductor two-phase LLC resonant converter.

of the auxiliary converter is actively controlled. This approach also reduces power and sampling losses through partial energy processing. Similarly, Wang et al. [14] achieves current sharing between two phases by introducing an auxiliary circuit along with a dedicated control strategy. [15] has proposed a topology using additional switches and capacitors for parallel *LLC* converters, which realizes current equalization parallel control. In [16], an interleaved *LLC* resonant converter utilizing phase-shift modulation is developed, and its optimal operating region is explored. The proposed PSM strategy compensates for current imbalance due to parameter mismatches. In [17], a new multiphase interleaved *LLC* resonant converter design is proposed. It employs three parallel *LLC* modules with star-connected transformer windings, significantly reducing output current ripple. Closed-loop phase-shift control is used to compensate current mismatch and achieve module current balance. Active current-sharing and hybrid methods, studied in [18], [19], [20], [21], use closed-loop control to dynamically adjust current in parallel modules, improving reliability and performance under varying loads. However, these methods typically depend on complex control schemes, multiple sensors, or auxiliary converters for accurate current sharing, which increases system complexity, introduces failure points, and raises material and manufacturing cost.

A class of passive current-sharing methods for multiphase *LLC* resonant converters is proposed in [22], [23], [24], [25], and [26], where current sharing is achieved by introducing coupled impedance, either to the resonant capacitor or resonant inductor between different phases, as illustrated in Fig. 2. When the resonant parameters are known, effective current sharing can be realized by adding either inductive or capacitive coupling impedance between phases. Even when the resonant parameters are unknown, introducing an infinitesimal coupling impedance between phases can still improve current-sharing performance. The primary advantage of these passive methods is their ability to achieve effective current sharing across the full load range without the need for additional control strategies or extra cost. Moreover, this passive current-sharing approach supports modular design of *LLC* resonant converters, which is particularly important for high-power applications.

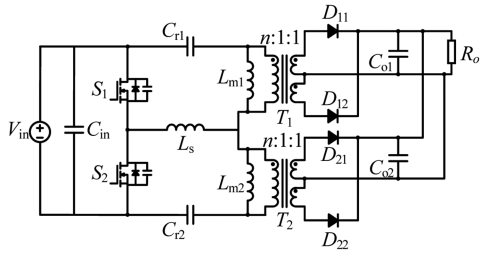


Fig. 3. Coupled half-bridge *LLC* resonant converter topology with reduced switches and current sharing ability.

In this article, a novel coupled half-bridge *LLC* resonant converter is proposed, which results in fewer power switches and current-sharing ability compared with previous methods. In addition, the two *LLC* resonant converters share the same resonant inductor, so the number of inductors is reduced. It helps to reduce costs and increase power density.

This rest of this article is organized as follows. Section II has introduced the derived coupled half-bridge *LLC* resonant converter and the operation modes are analyzed. In Section III, the effect of different resonant parameters on the current sharing error is analyzed, and the current sharing effect of the proposed *LLC* resonant converter is analyzed in the case of the worst parameter error, followed by simulation and experimental results in Section IV. Finally, Section V concludes this article.

II. Derived Coupled Topology

A. Topology Structure

The coupled half-bridge *LLC* resonant converter topology is shown in Fig. 3. It can be seen that the two *LLC* resonant converters share same two power switches S_1 and S_2, and there are only two switch devices compared to conventional paralleled *LLC* resonant converters. In addition, the two resonant networks share the same resonant inductor L_s, which is obtained by simplifying the common resonant inductor passive current sharing topology. Apart from that, the first resonant network contains resonant capacitor C_{r1} and magnetizing inductor L_{m1}, and the second resonant network contains resonant capacitor C_{r2} and magnetizing inductor L_{m2}.Transformers and rectifier circuits are same with traditional two-phase *LLC* resonant converters.

B. Operation Modes

In order to analyze operation mode of the coupled *LLC* resonant converter, some assumptions need to be made.

1) Switches S_1 and S_2 have the same parasitic capacitor.
2) Resonant capacitor $C_{r1} = C_{r2} = C_r$.
3) Magnetizing inductor $L_{m1} = L_{m2} = L_m$.

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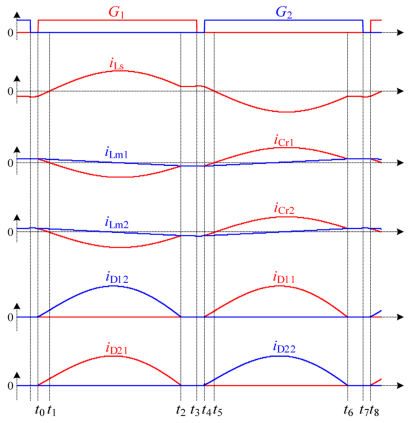


Fig. 4. Key waveforms of the coupled half-bridge *LLC* resonant converter.

4) Transformers T_1 and T_2 are identical and turn ratios are both n.

Based on the assumptions, it can be seen that the equivalent inductive value of the each resonant network is twice that of the resonant inductor $L_{s,}$ and the resonant frequency can be calculated for each *LLC* resonant converter unit as follows:

$$f_r = \frac{1}{2\pi\sqrt{2L_sC_r}}. \tag{1}$$

There are eight operation modes of the proposed coupled half-bridge *LLC* resonant converter during a switching period, and the key waveforms are shown in Fig. 4. Further, the detailed operation modes are shown in Fig. 5.

Mode 1 (t_0, t_1): This mode starts at t_0, at this time, switch S_1 is ON and S_2 is OFF. At t_0, the current of common inductor L_s is negative, so the parasitic capacitance of switch S_1 is discharged and the parasitic capacitance of switch S_2 is charged, the body diode of switch S_1 is ON. Therefore, ZVS performance of S_1 is achieved. The diodes D_{12} and D_{21} are ON to flow rectifier current, and it can be seen that the voltage of magnetizing inductors is clamped by -n times the output voltage. Therefore, the currents of magnetizing inductors i_{Lm1} and i_{Lm2} decreased linearly. Finally, the power is transferred from input voltage V_{in} to output load R_o in this mode.

Mode 2 (t_1, t_2): In this mode, switch S_1 is ON and S_2 is OFF. At t_1, the current of common inductor, resonant capacitance C_{r1} and resonant capacitance C_{r2} is equal to zero, and mode 2 is started. The voltage stress of switch S_2 is equal to V_{in}, and the current of common inductor L_s is positive after t_1. The diodes D_{12} and D_{21} keep ON, and the voltage of magnetizing inductors is still clamped by -n times the output voltage. Therefore, the currents of magnetizing inductors i_{Lm1} and i_{Lm2} decreased linearly. The power is transferred from input voltage V_{in} to output load R_o in this mode, and this mode ends when $i_{Cr1} = i_{Lm1}$ and $i_{Cr2} = i_{Lm2}$.

Mode 3 (t_2, t_3): Switch S_1 keep ON and S_2 keep OFF. The current of diodes D_{12} and D_{21} become zero, meanwhile, $i_{Cr1} = i_{Lm1}$ and $i_{Cr2} = i_{Lm2}$. There is no power transmitted from the primary side to secondary side of transformer. The voltage of magnetizing inductors is no longer clamped by output voltage. At this time, resonance is achieved in primary side by three devices, and the resonant current flows into the magnetizing inductor. Besides, the output power is supplied by output capacitors C_{o1} and C_{o2}.

Mode 4 (t_3, t_4): In this time period, both switch S_1 and S_2 are OFF, and it is called dead time. At $t = t_3$, i_{Ls} is larger than zero, as a result, the parasitic capacitor of S_1 is charged and the parasitic capacitor of S_2 is discharged, and the body diode of S_2 is on when the drain-source voltage of S_2 decreased to zero, which provides the conditions for the realization of the soft switch.

Mode 5 (t_4, t_5): This mode starts at t_4, at this time, switch S_1 is OFF and S_2 is ON. Since the voltage across the switch S_2 dropped to 0 in the previous mode and the body diode had already been conducting, the turn-ON process of switch S_2 achieved ZVS. At t_4, the current of common inductor L_s is negative and will descend to zero at t_5. The diodes D_{12} and D_{21} are ON to flow rectifier current, and it can be seen that the voltage of magnetizing inductors is clamped by n times the output voltage. Therefore, the currents of magnetizing inductors i_{Lm1} and i_{Lm2} increased linearly. Finally, the power is transferred from input voltage V_{in} to output load R_o in this mode.

Mode 6 (t_5, t_6): In this mode, switch S_1 is OFF and S_2 is ON. At t_5, the current of common inductor, resonant capacitance C_{r1} and C_{r2} is equal to zero. The voltage stress of switch S_1 is equal to V_{in}, and the current of common inductor L_{s} is negative after t_5. The diodes D_{12} and D_{21} keep ON, and the voltage of magnetizing inductors is still clamped by n times the output voltage. Therefore, the currents of magnetizing inductors i_{Lm1} and i_{Lm2} increased linearly. The power is transferred from input voltage V_{in} to output load R_{o} in this mode, and this mode ends when $i_{Cr1} = i_{Lm1}$ and $i_{Cr2} = i_{Lm2}$.

Mode 7 (t_6, t_7): Switch S1 keep OFF and S_2 keep ON. The current of diodes D_{12} and D_{21} become zero, meanwhile, $i_{Cr1} = i_{Lm1}$ and $i_{Cr2} = i_{Lm2}$. There is no power transmitted from the primary side to secondary side of transformer. The voltage of magnetizing inductors is no longer clamped by output voltage. At this time, resonance is achieved in primary side by three devices, and the resonant current flows into the magnetizing inductor. Besides, the output power is supplied by output capacitors C_{o1} and C_{o2}.

Mode 8 (t_7, t_8): In this mode, both switch S_1 and S_2 are OFF, and it is called dead time. At t_7, i_{Ls} is less than zero, as a result, the parasitic capacitor of S_1 is discharged and the parasitic capacitor of S_2 is charged, and the body diode of S_1 is ON when the drain-source voltage of S_1 decreased to zero, which provides the conditions for the realization of the soft switch.

III. Analysis for Current Sharing Effect

Wang et al. [23] provide a useful analytical basis for understanding the current-sharing mechanism of resonant converters under parameter mismatches. Building on this idea, the proposed topology is analyzed in this article using a similar framework. Since the proposed topology differs in circuit configuration and operating characteristics, a concise analytical formulation is presented here to explicitly demonstrate its automatic current-sharing behavior and to validate its performance under resonant-tank parameter variations.

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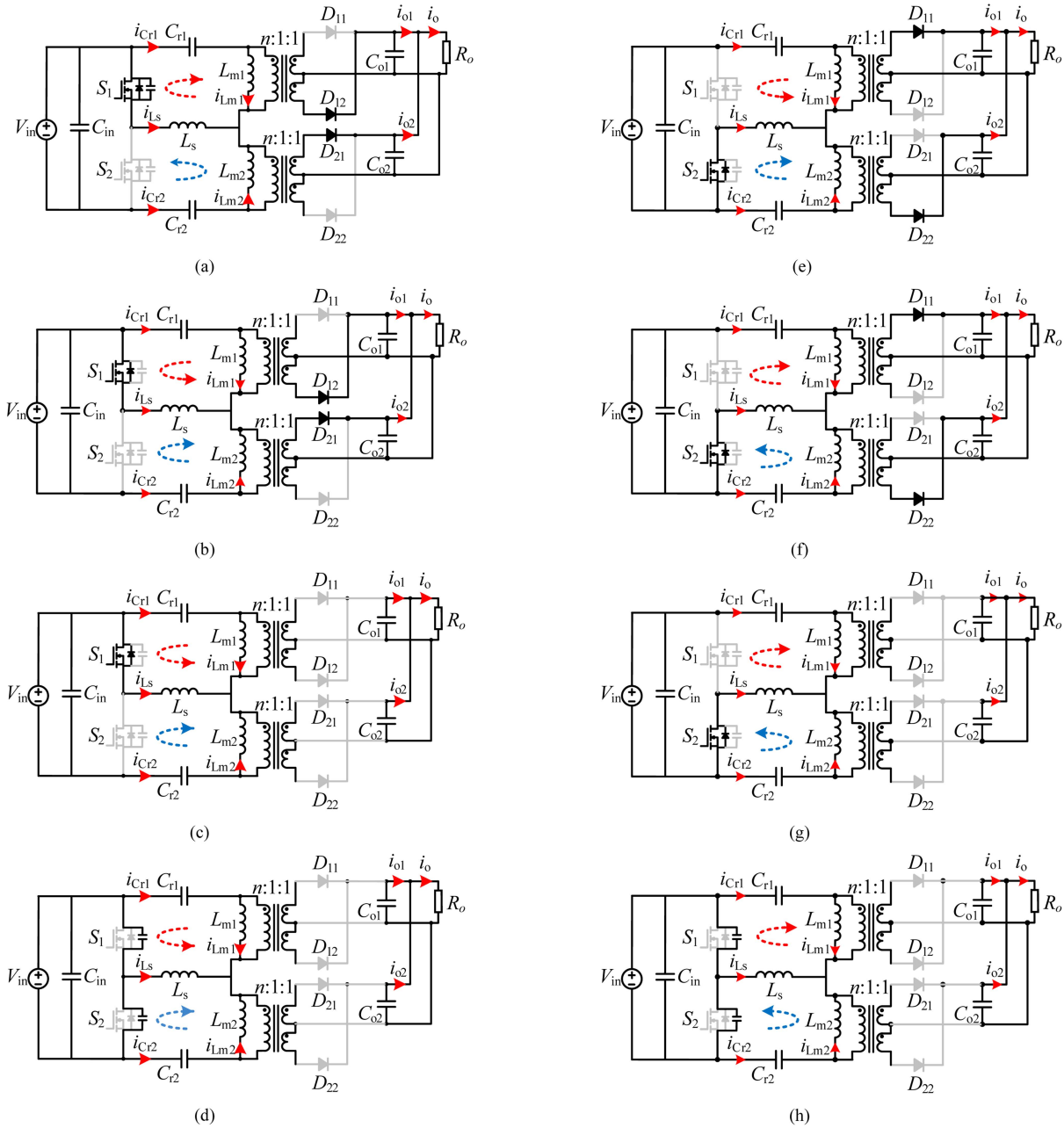


Fig. 5. Operation modes of the proposed coupled half-bridge *LLC* resonant converter. (a) Mode 1. (b) Mode 2. (c) Mode 3. (d) Mode 4. (e) Mode 5. (f) Mode 6. (g) Mode 7. (h) Mode 8.

In accordance with the proposed coupled half-bridge *LLC* resonant converter, its equivalent circuit can be derived, comprising circuits 1 and 2 as illustrated in Fig. 6. Here, the load R_o is assumed to be partitioned into R_{o1} and R_{o2}, with their respective equivalent load resistances denoted as R_{ac1} and R_{ac2}. Additionally, both circuits share a common resonant inductor L_s, thereby facilitating the establishment of an impedance relationship between them.

Before the establishment of the model, it is essential to define the existing parameters, which include both the parameters and their associated errors for the resonant capacitor and magnetizing inductance, as illustrated in (2). Among them, the parameters b and c represent the size of the error. The topology shares a common resonant inductor *Ls*, so there is no parameter difference in the resonant inductor

$$\begin{cases} C_{r1} = C_r, C_{r2} = bC_r \\ L_{m1} = L_m, L_{m2} = cL_m \\ Z_{Cr} = 1/sC_r, Z_{Lm} = sL_m \end{cases}. \tag{2}$$

Suppose the output power of circuit 1 as P_{o1} and that of circuit 2 as P_{o2}, with the total output power represented by P_{o}. If we define the output voltage across both paths as V_{o}, then the expressions for load and output power can be articulated as follow:

$$\begin{cases} R_{\mathrm{o1}} = \frac{{V_{\mathrm{o}}}^2}{P_{\mathrm{o1}}} \\ P_{\mathrm{o}} = P_{\mathrm{o1}} + P_{\mathrm{o2}} \\ R_{\mathrm{o2}} = \frac{{V_{\mathrm{o}}}^2}{P_{\mathrm{o2}}} \end{cases}. \tag{3}$$

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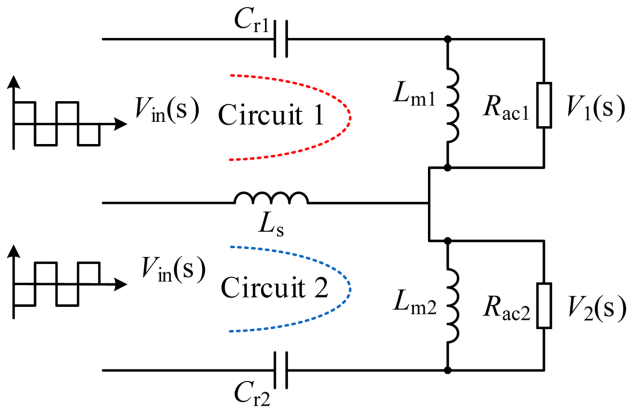


Fig. 6. Equivalent circuit of the coupled half-bridge *LLC* resonant converter.

The coefficient k serves to quantify the power differential between the two circuits, with its value constrained to the range of 0 to 1. A value of k equal to 0 indicates that all power is allocated to circuit 2, whereas a value of k equal to 1 signifies that all power is directed towards circuit 1

$$\begin{cases} P_{o1} = kP_o, \\ P_{o2} = (1-k)P_o \end{cases}. \tag{4}$$

In accordance with the principles of first harmonic approximation, we can derive the following expression:

$$\begin{cases} R_{o1} = \frac{1}{k}R_o, R_{o2} = \frac{1}{1-k}R_o \\ R_{\mathrm{ac1}} = \frac{8n^2}{\pi^2}R_{o1}, R_{\mathrm{ac2}} = \frac{8n^2}{\pi^2}R_{o2} \end{cases}. \tag{5}$$

According to the relationship of parallel connection on the output side of the topology in this article, the following formula can be obtained based on the principle that the voltage on the output load is equal:

$$|V_1(s)| = |V_2(s)|. \tag{6}$$

Based on Fig. 6, the corresponding transfer function expression can be established according to the relationship of impedance, and the results are shown in the following formula:

$$\begin{cases} \frac{|V_1(s)|}{|V_{\mathrm{in}}(s)|} = \left| \frac{R_{\mathrm{ac1}}//Z_{Lm}}{R_{\mathrm{ac1}}//Z_{Lm}+Z_{Ls}+Z_{Cr}} \right| \\ \frac{|V_2(s)|}{|V_{\mathrm{in}}(s)|} = \left| \frac{R_{\mathrm{ac2}}//cZ_{Lm}}{R_{\mathrm{ac2}}//cZ_{Lm}+Z_{Ls}+Z_{Cr}/b} \right| \end{cases}. \tag{7}$$

By integrating (2), (4), (5), (6), and (7), a parametric equation for the coefficient k can be formulated, as delineated in (8). The coefficients A, B, and C within this parametric equation are articulated in

$$Ak^2 + Bk + C = 0 \tag{8}$$

$$\begin{cases} A = -b^2c^2(-C_rL_mL_r\omega^3 + L_m\omega)^2\pi^4 \\ \quad + c^2{L_m}^2\omega^2(C_rL_rb\omega^2 - 1)^2\pi^4 \\ B = -2c^2{L_m}^2\omega^2(C_rL_rb\omega^2 - 1)^2\pi^4 \\ C = c^2{L_m}^2\omega^2(C_rL_rb\omega^2 - 1)^2\pi^4 + \\ 64n^4(R_o - bC_r(L_r + L_mc)\omega^2R_o)^2 \\ -64b^2c^2n^4(-1 + C_r(L_r + L_m)\omega^2)^2{R_o}^2 \end{cases}. \tag{9}$$

In addition, by analyzing the relationship between output power and output current, the relationship between current distribution error and coefficient k can be obtained, I_1 and I_2 represent the output currents of circuit 1 and circuit 2. The value of the current sharing error σ can reflect the current sharing effect of the proposed converter. The smaller the value of σ, the better the uniform flow effect

$$\sigma = \left| \frac{I_1 - I_2}{I_1 + I_2} \right| = |1 - 2k|. \tag{10}$$

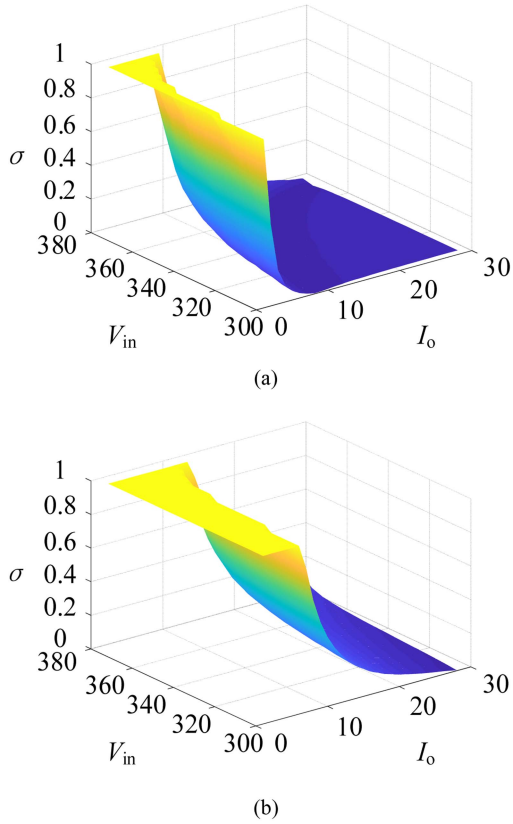


Fig. 7. Current sharing error σ of the proposed *LLC* converter under different input voltage V_{in} and different output current I_o. (a) b=1.05, c=0.95. (b) b=1.05, c=1.05.

Utilizing the aforementioned mathematical model, the current sharing ratio of the proposed converter can be calculated under different operating conditions. The relationship between the output current sharing error and the input voltage and output current can be obtained, as illustrated in Fig. 7.

Fig. 7(a) analyzes the case where the resonant capacitor parameters differ by 1.05 and the magnetic inductance parameters differ by 0.95 ($b = 1.05$, $c = 0.95$). Under this set of parameter errors, the proposed converter has good current sharing performance in most of the load range, but poor current sharing performance in very light load. Poor current sharing performance in very light load is an inherent problem, and the requirement for current sharing in this load condition will not be high. Fig. 7(b) analyzes the case where the resonant capacitor parameters differ by 1.05 and the magnetic inductance parameters differ by 1.05 ($b = 1.05$ and $c = 1.05$). When the parameter errors deviate in the same direction, there will be larger current sharing errors. The current sharing performance in most of the load range is worse than that of the previous set of parameters, but with the increase

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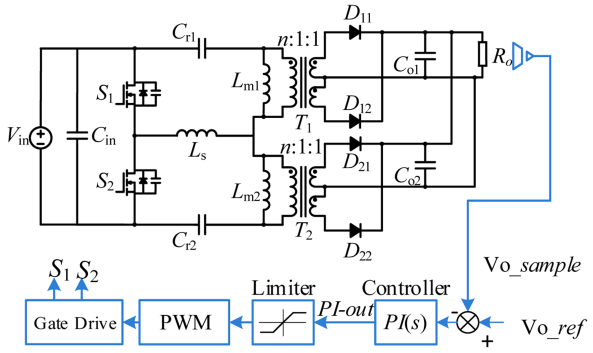


Fig. 8. Control scheme of the proposed coupled half-bridge *LLC* resonant converter.

of load, good current sharing performance can be achieved in this set of parameters. Better current sharing performance is achieved in light load compared with traditional topology.

Furthermore, the proposed coupled half-bridge *LLC* resonant converter facilitates automatic current sharing through a shared inductor L_s. Thus, varying coupling impedances cannot be introduced to enhance current-sharing efficacy for different parameter deviations yet still satisfy required currents within specified load ranges.

IV. Simulation and Experimental Results

To verify the correctness and effectiveness of the current-sharing capability of the proposed coupled half-bridge *LLC* resonant converter, both simulation and experimental validations are necessary. A simulation platform was established in PSIM for simulation verification, while experimental validation was conducted on a hardware platform.

In both simulations and experimental setups, the two parallel *LLC* resonant converters employ conventional pulse frequency modulation techniques, as illustrated in Fig. 8. In the output-parallel configuration, the output voltages of both modules are identical; therefore, it is sufficient to sample the output voltage of only one module. The error signal is derived by subtracting the measured output voltage from the reference voltage. This error is then processed by a proportional-integral (PI) controller, which generates a control signal. The output of the PI controller is subsequently passed through a limiter before being fed to the PFM generator, which produces the corresponding drive signal.

A. Simulation Validations

Table I gives the simulation parameters of the proposed topology, which are consistent with the design outlined in the previous section. To verify the correctness and effectiveness of the current-sharing capability of the proposed coupled half-bridge *LLC* resonant converter, parameter mismatches were intentionally introduced: the values of the two resonant capacitors were increased by a factor of 1.05, and similarly, the magnetizing inductance was also increased by a factor of 1.05.

Fig. 9 shows the simulation waveform of the proposed topology under 5% parameter error. Fig. 9(a) shows the current

TABLE I
Simulation and Prototype Parameters

Category	Value
Input voltage V_{dc}	340V
Output voltage V_{out}	48V
Output power P	1.0kW
Resonant frequency f_r	90kHz
Resonant capacitor C_{r1}/C_{r2}	178nF/178*1.05nF
Resonant inductor L_s	8.8uH
Magnetizing inductor L_{m1}/L_{m2}	270uH/270*1.05uH
Transformer ration n	18:5:5

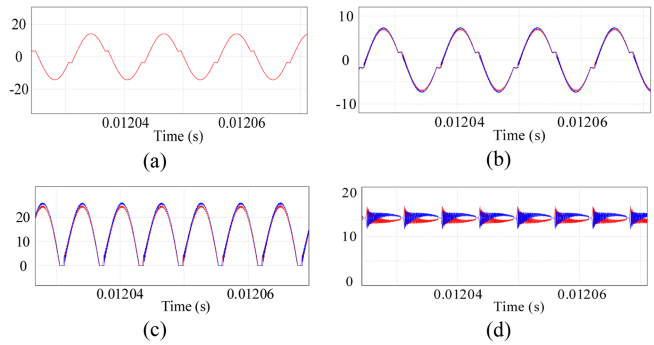


Fig. 9. Simulation waveform of the proposed topology under 5% parameter error. (a) Waveform of public inductance *Ls*. (b) Resonant current waveform. (c) Rectifier output waveform. (d) Output current waveform.

waveform on the common inductance L_s, which is a quasi-sinusoidal waveform and is related to the working frequency at this time. It can be seen that it is consistent with the theoretical analysis in the previous paper. Fig. 9(b) shows the current waveform on the two resonant capacitors. It can be seen from the waveform that the two current waveforms are basically coincident at this time, and there is only a small current error, so the current can be equalized. Fig. 9(c) shows the rectified waveforms of the two rectifier bridges on the output side. The waveforms are basically consistent, with only a small current difference at the peak. The final output current waveform is shown in Fig. 9(d). It can be seen that the average current is basically equal at this time, and there are only slight current differences. Therefore, the proposed coupled half-bridge *LLC* resonant converter can achieve good current balance.

Figs. 10 and 11 show the transient simulation waveforms of the proposed coupled half-bridge *LLC* resonant converter under 5% parameter error. Simulations were conducted for both half-load to full-load switching and full-load to half-load switching. Figs. 10 and 11(a) show the resonant current waveform and Figs. 10 and 11(b) present the rectifier output waveform. During transient states, both resonant current and rectifier output current achieve excellent current balance.

B. Experimental Validations

To verify the current-sharing capability of the proposed topology, an experimental platform was constructed, as shown in Fig. 12. Due to the difficulty in ensuring that the hardware

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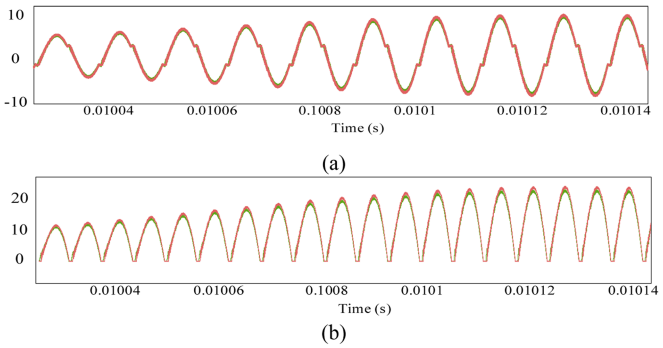


Fig. 10. Transient waveform of topology switching from half-load to full-load with 5% parameter error. (a) Resonant current waveform. (b) Rectifier output waveform.

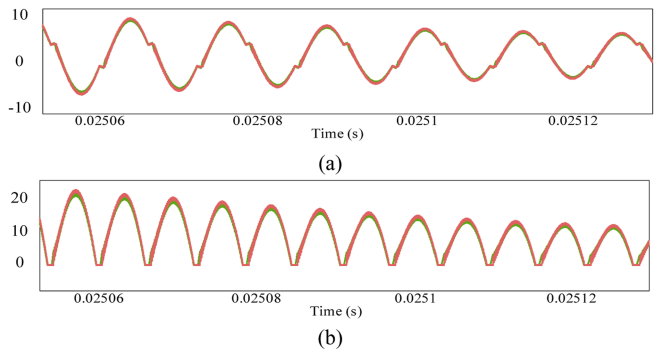


Fig. 11. Transient waveform of topology switching from half-load to full-load with 5% parameter error. (a) Resonant current waveform. (b) Rectifier output waveform.

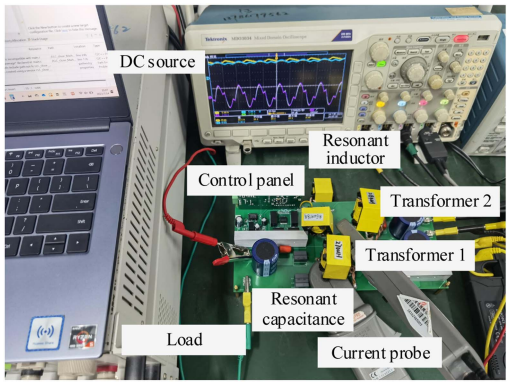


Fig. 12. Experiment platform of the coupled half-bridge *LLC* resonant converter.

parameters exactly match the theoretical values, some discrepancies exist in the resonant parameters. The specific experimental parameters are given in Table II.

The results of the steady-state experiment are shown in Fig. 13. Fig. 13(a) shows the experimental waveform at 20% load, when the output current of the two channels is 2.23 and 2.02 A, and the current error is 4.94%. Fig. 13(b) shows the experimental waveform at 50% load, when the output current of the two channels is 5.19 and 4.86 A respectively, and the current

TABLE II
RESONANT PARAMETERS OF THE HARDWARE PLATFORM

Category	Value
Input voltage V_{dc}	340V
Output voltage V_{out}	48V
Output power P	1.0kW
Resonant frequency f_r	90kHz
Resonant capacitor C_{r1}/C_{r2}	177nF/189nF
Resonant inductor Ls	9uH
Magnetizing inductor L_{m1}/L_{m2}	270uH/284uH
Transformer ration n	18:5:5
Dead time td	300ns
Junction capacitance Coss	200pF

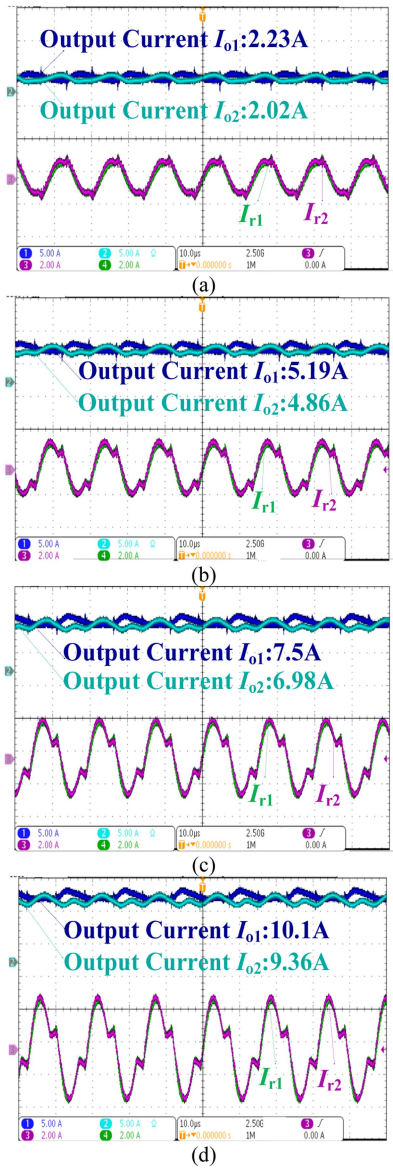


Fig. 13. Experimental results of the coupled half-bridge *LLC* resonant converter in different load. (a) 20% load. (b) 50% load. (c) 75% load. (d) 100% load.

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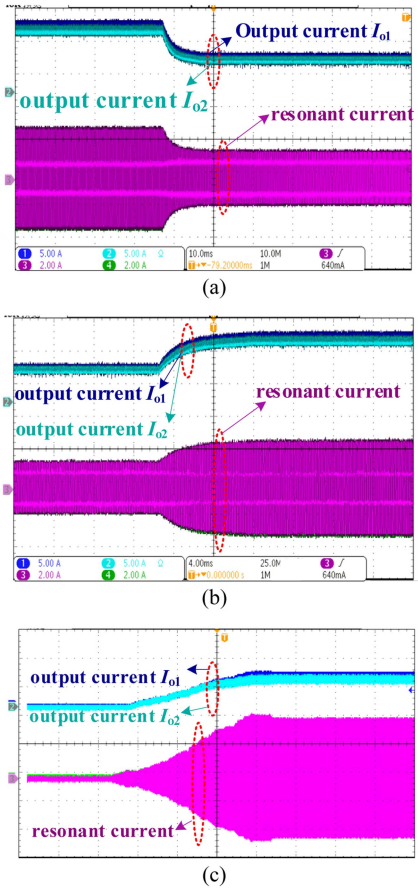


Fig. 14. Dynamic experimental results of the coupled half-bridge *LLC* resonant converter. (a) Full-load to half-load transition. (b) Half-load to full-load transition. (c) Startup transient.

error is 3.28%. Fig. 13(c) shows the experimental waveform at a 75% load, where the output currents for the two paths are 7.50 and 6.98 A, with an average current error of 3.59%. Fig. 13(d) shows the experimental waveform at a 100% load, where the output currents for the two paths are 10.1 and 9.36 A, with an average current error of 3.8%. Compared with traditional solutions, this topology maintains a current sharing error within 5% without requiring active control strategies.

To further verify the reliability of the coupled half-bridge *LLC* resonant converter, the dynamic experiments were conducted as shown in Fig. 14. In Fig. 14(a), the dynamic experimental waveform during the transition from full load to half load is illustrated. Similarly, in Fig. 14(b), the waveform during the transition from half load to full load is shown. Fig. 14(c) shows the waveforms during the startup process. The dynamic experiments demonstrate that the proposed topology exhibits excellent dynamic response and current sharing characteristics.

Overall, the proposed the coupled half-bridge *LLC* resonant converter can achieve automatic current sharing across all load ranges and demonstrates excellent current sharing performance.

Subsequently, experimental verification of soft switching was performed once more, including light-load tests at 5% and 10% of the rated power. The voltage waveforms across the half-bridge and the current waveforms through the resonant inductor were

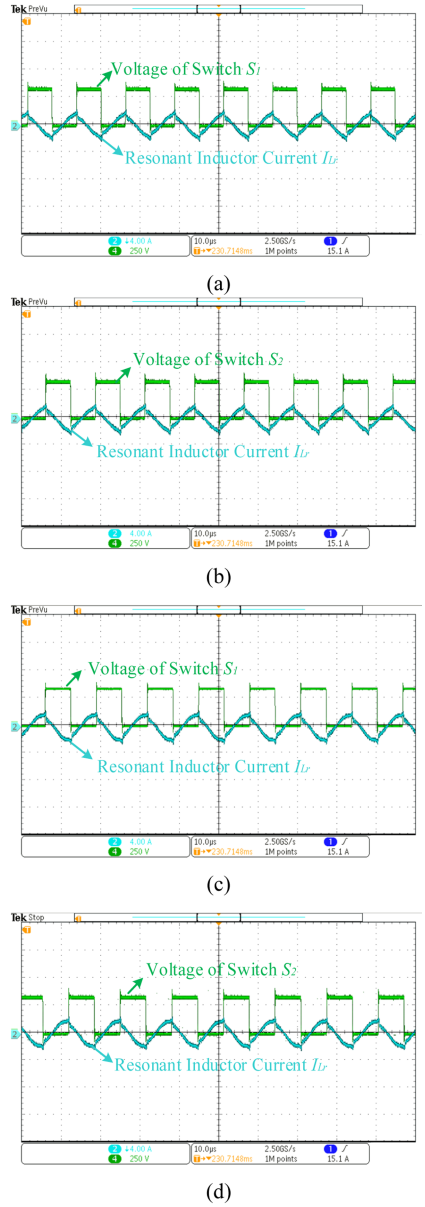


Fig. 15. Voltage and resonant inductance current waveform diagram of switching tube under light load. (a) and (b) Soft-switching experimental waveforms of switches $S1$ and $S2$ at 50 W. (c) and (d) Soft-switching experimental waveforms of switches $S1$ and $S2$ at 100 W.

separately measured. During the dead time prior to the switching transistor's turn-on, the current from the resonant inductor charged and discharged the parasitic capacitance of the switching transistor. The results confirm that the converter is capable of achieving soft switching even under light-load conditions. The following experimental waveforms clearly illustrate this capability in Fig. 15.

C. *Efficiency Tests*

Finally, efficiency tests and loss calculations were conducted. Efficiency tests were conducted across varying load conditions: 10%; 30%; 50%; 80%; and 100%. The measured efficiencies stand at 88.2%, 95.1%, 96.3%, 97.1%, and 96.4%, respectively.

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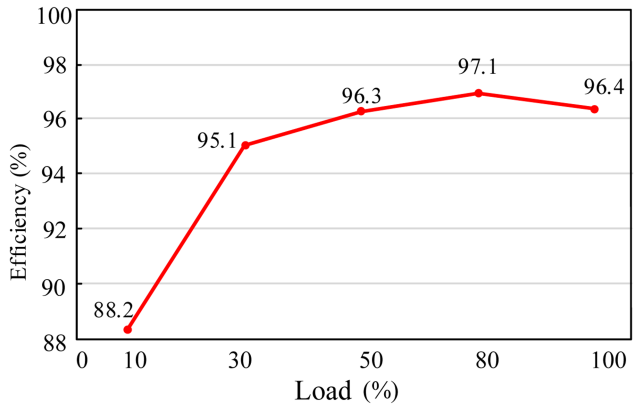


Fig. 16. Curve of change in efficiency of the coupled half-bridge *LLC* resonant converter.

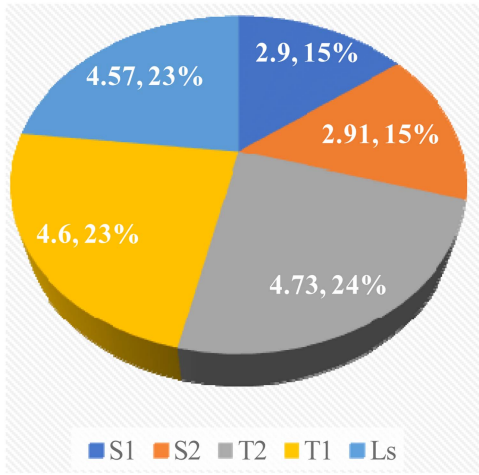


Fig. 17. Main loss distribution pattern.

Notably, the proposed topology achieves a remarkable peak efficiency of 97.1% at 80% load and maintains an impressive full-load efficiency of 96.4%. The resulting efficiency curve is presented in Fig. 16.

The efficiency trend is consistent with the operating characteristics of the converter. Under light load conditions, the proportion of fixed losses, such as gate driving loss, magnetic core loss, and control circuit loss, becomes relatively high compared with the output power. Therefore, the efficiency is relatively low under light loads. As the load increases, the proportion of these fixed losses decreases and the efficiency improves significantly. At full load, the efficiency decreases compared with the peak point. This is mainly due to increased current stress leading to higher losses in power switches, rectifier devices, and transformer windings.

In LLC resonant converters, power losses mainly arise from MOSFETs, rectifier diodes, inductance, capacitors and transformers. Since the primary side achieves zero-voltage switching during operation, turn-ON losses of primary side switching devices are negligible. Therefore, the main concerns are turn-OFF loss and conduction losses. Therefore, the power loss of switching devices under rated operating conditions can be calculated using the following two equations:

$$P_{\text{sw1_loss}} = {I_{S_1,\text{rms}}}^2 R_{ds} + f_s[E_{\text{off}}(I_{\text{off}}, V_{\text{in}}) - E_{\text{off}}(0, V_{\text{in}})] = 2.90 \text{ W} \quad (11)$$

$$P_{\text{sw2_loss}} = {I_{S_2,\text{rms}}}^2 R_{ds} + f_s[E_{\text{off}}(I_{\text{off}}, V_{\text{in}}) - E_{\text{off}}(0, V_{\text{in}})] = 2.91 \text{ W} \quad (12)$$

where *I*rms is the effective current, and *R*ds is the ON-resistance of the switching device. *E*off represents the turn-OFF loss. The calculated losses of the two primary side MOSFETs are about 2.90 and 2.91 W, corresponding to a total switching-device loss of about 5.81 W at full load. Moreover, the similar loss values of the two primary side MOSFETs also indicate good current sharing.

For magnetic components, energy losses mainly consist of copper losses and iron losses. The internal resistance *RLr* of a resonant inductor is 0.0528 Ω. The transformers exhibit primary side resistance *R*pri = 0.0270 Ω and secondary side resistance *R*sec = 0.0167 Ω. Core losses are predominantly determined by switching frequency and core material properties. The relevant calculation equations are as follows.

Resonant inductance loss can be calculated using the following equation:

$$P_{\text{loss_}Lr} = R_{Lr} I_{\text{rms}}^2 + V_{e_Lr} k f_s^{\alpha_{\text{core}}} \Delta B_{m_\text{Lr}}^{\beta_{\text{core}}} = 4.57 \text{ W}. \quad (13)$$

Transformer losses can be calculated using the following equation:

$$P_{\text{loss_Trans1_pri}} = I_{\text{1_pri_rms}}^2 R_{\text{1_pri}} + I_{\text{1_sec_rms}}^2 R_{\text{1_sec}} + V_{\text{e_Trans}} k f_{\text{s}}^{\alpha_{\text{core}}} \Delta B_{\text{1_m_Trans}}^{\beta_{\text{core}}} = 4.60 \text{ W} \quad (14)$$

$$P_{\text{loss_Trans2_pri}} = I_{\text{2_pri_rms}}^2 R_{\text{2_pri}} + I_{\text{2_sec_rms}}^2 R_{\text{2_sec}} + V_{\text{e_Trans}} k f_s^{\alpha_{\text{core}}} \Delta B_{\text{2_m_Trans}}^{\beta_{\text{core}}} = 4.73 \text{ W}. \quad (15)$$

ΔBm represents flux swing, where k, α, and β are Steinmetz coefficients. Specific parameters are obtained from the manufacturer. Due to the use of the same magnetic cores in transformers, the core volume and Steinmetz coefficients remain consistent. For the 1 kW prototype, the calculated transformer losses are about 4.60 and 4.73 W for the two transformers, corresponding to a total transformer loss of about 9.33 W. This accounts for only 0.93% of the output power, which is considered reasonable for a high-frequency *LLC* converter. Moreover, the close loss values of the two transformers also mean good current sharing.

According to the calculated results, the loss distribution of two transformers, two switching transistors, and resonant inductors is shown in Fig. 17.

V. Conclusion

This article presents a coupled half-bridge *LLC* resonant converter. Compared with traditional parallel half-bridge resonant converters, this design achieves superior performance with reduced switching elements while maintaining moderate power

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output. Analysis, simulation, and experimental results demonstrate that under worst-case conditions, the current sharing error between the two circuits remains below 5%. When employing the proposed *LLC* converter, the resonant current error is only 2.5% at full rated load power.

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